Binary-SNGP: Uncertainty-Aware TinyML Anomaly Detection on FPGA for Edge AI in Industrial IoT

The Tuan Trinh
Faculty of Electrical and Electronic Engineering
PHENIKAA School of Engineering
PHENIKAA University
Hanoi 12116, Vietnam
tuan.trinhthe@phenikaa-uni.edu.vn

Nguyen Du
ICSLab
PHENIKAA School of Engineering
PHENIKAA University
Hanoi 12116, Vietnam
ngdu1212@gmail.com

Khoa Nguyen Dang
Faculty of Engineering and Technology
International School
Vietnam National University
Hanoi, Vietnam
khoand@vnuis.edu.vn

Xuanque Nguyen*
Faculty of Information System
PHENIKAA School of Computing
PHENIKAA University
Hanoi 12116, Vietnam
*Corresponding: *que.nguyenxuan@phenikaa-uni.edu.vn*

***Abstract*—Industrial Internet of Things (IIoT) edge nodes that classify machine vibration fail silently when their input drifts from the training data, and ensemble or Monte-Carlo-dropout uncertainty estimators multiply the hardware cost. This paper presents Binary-SNGP, a single-pass, distance-aware uncertainty model for Edge AI, co-designed for low-cost FPGAs with a TinyML footprint of 35.9 kbit. A 1-bit convolutional backbone feeds a Gaussian-process output layer reformulated for integer hardware: ± 1 random features with a 1,024-entry cosine ROM, a rank-16 variance that provably never under-estimates the exact variance, and a decision made by two comparators. On a Cyclone IV, the accelerator matches its integer reference model on 2,282 recorded windows and needs a fixed 1,778 cycles (35.6 μs at 50 MHz) per window. On the same backbone (five seeds), it reaches a mean out-of-distribution AUROC of 0.98, against 0.79 for a deep ensemble and 0.62 for MC dropout. In a field test, a sensor-mounting shift made the classifier report every window as abnormal, and Binary-SNGP flagged 99.4% of these windows as unknown (deep ensemble 84%), and an orientation-robust variant removes the failure. The node sends 12 bytes per window.**

***Index Terms*—Industrial Internet of Things, Edge AI, TinyML, Binarized Neural Networks, FPGA Accelerator, Uncertainty Estimation, Spectral-normalized Neural Gaussian Process, Out-of-Distribution Detection, Predictive Maintenance**

I. Introduction

In the Industrial Internet of Things (IIoT), condition monitoring of rotating machines increasingly runs on the sensor node itself, which must meet real-time, energy and bandwidth constraints while classifying vibration from a low-cost MEMS inertial sensor without sending raw data to the cloud [1]. This move of inference from the cloud to the device, known as Edge AI [2], is pushed to its limit by TinyML, which runs models within kilobytes of memory and milliwatt budgets and lists anomaly detection among its benchmark tasks [3]. One-dimensional convolutional networks are a standard tool for this task because they learn directly from the raw signal [4]. FPGAs suit such monitors well: they give deterministic, cycle-level latency, they host the sensor interface and the actuator control in the same fabric, and binarized neural networks (BNNs) replace multiplications by additions or XNOR–popcount operations that map onto look-up tables [5]–[7]. A deployed monitor, however, inevitably meets inputs that its training set never contained, such as a re-mounted sensor, a different machine or a failing channel.

A standard classifier returns a confident label for such inputs, and its softmax confidence is a poor indicator of the failure [8], [9]. In our own field test a correctly verified FPGA classifier reported every window of a healthy fan as abnormal after the sensor had been re-mounted. Predictive uncertainty is the established remedy, also in prognostics and health management [10], but Monte-Carlo dropout [11] and deep ensembles [12] obtain it from repeated passes or several networks, so FPGA accelerators for Bayesian networks pay with latency, memory and on-chip random number generators [13], [14]. Distance-aware single-pass methods avoid this cost: the Spectral-normalized Neural Gaussian Process (SNGP) [15] replaces the output layer by a Gaussian process (GP) approximated with random features [16], and DDU [17] fits a feature density. SNGP, however, requires a dense floating-point covariance matrix and trigonometric functions, and, to our knowledge, it has not been combined with a binarized backbone in hardware.

This paper presents *Binary-SNGP*, a binary-backbone GP model co-designed with its FPGA datapath. The contributions are:

(a) **Integer GP head.** Random features with ± 1 weights and a cosine ROM, and a rank-R variance whose truncation provably never under-estimates the exact variance, so that the decision reduces to two comparators.

(b) **FPGA IIoT edge node.** A Cyclone IV accelerator with a

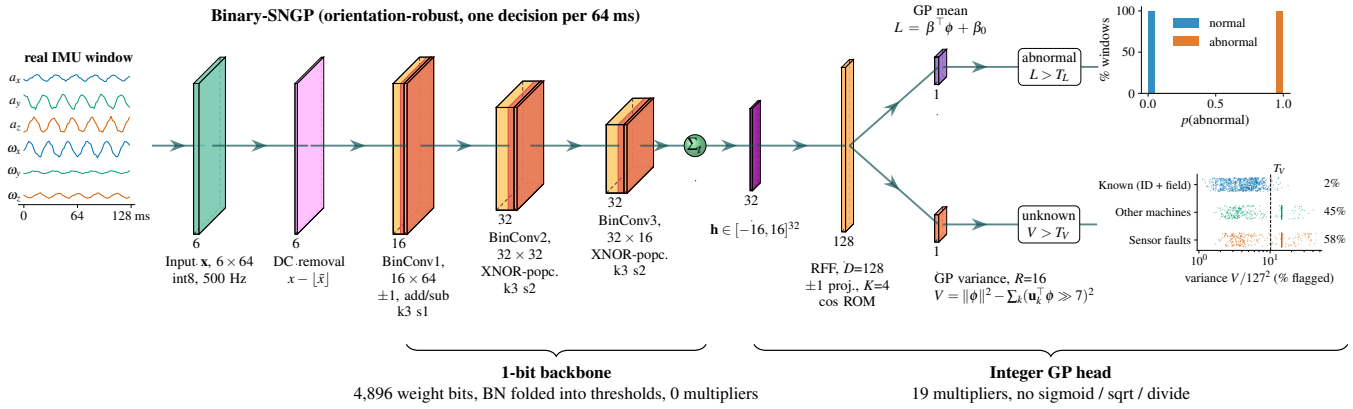


Figure 1. Binary-SNGP network architecture with integer layer sizes. Left: a real input window. Box height is proportional to sequence length and width to channels; the dark band is the folded batch-norm/sign threshold. Right: outputs of the two heads on held-out data, i.e. p(abnormal) of the mean head and the variance V with the share of windows above T_V.

closed-form cycle model, verified bit for bit on the board, with a TinyML footprint of 35.9 kbit and a decision plus its uncertainty in a 12-byte packet.

(c) **Fair comparison.** MC dropout, deep ensembles, Mahalanobis distance and Binary-SNGP on the same binary backbone over five seeds, including their hardware cost.

(d) **Field case study.** A sensor-mounting shift exposed by the uncertainty output alone, then removed by an orientation-robust variant trained only on the existing recordings.

The complete hardware RTL codebase, integer reference model, and evaluation scripts are publicly available at https://github.com/thetuantrinh/FPGA-Binary-SNGP-Anomaly-Detection.

II. Methodology

A. Input Coding and Binary Backbone

A 6-axis IMU (ADXL345 accelerometer, ITG3205 gyroscope) on the fan hub delivers $\mathbf{a}_t$ [g] and $\boldsymbol{\omega}_t$ [°/s] at $f_s = 500$ Hz. The model observes windows of $W = 64$ samples with a stride of 32, so a decision is due every $T_d = 64$ ms, and every channel i is mapped to a signed 8-bit code

$$x_{t,i} = \mathrm{clip}\big(\mathrm{round}(16\,(r_{t,i} - \mu_i)/\sigma_i),\, -127,\, 127\big), \qquad (1)$$

where $r_{t,i}$ is the raw reading. On the FPGA, the sensor scale, μ_i and σ_i fold into one multiply–add per channel. Three 1D convolutions (kernel 3, padding 1, strides 1, 2, 2) map $\mathbf{x} \in \mathbb{Z}^{6\times 64}$ to 16×64, 32×32 and 32×16 maps (Fig. 1). Weights are binarized with the sign function and trained with the straight-through estimator [5], [18], and activations are binarized after batch normalisation (BN) [19]. The pre-activation of layer ℓ and channel c is the integer

$$a_{c,t}^{(\ell)} = \textstyle\sum_{i,k} w_{c,i,k}^{(\ell)}\, z_{i,\,s_\ell t+k-1}^{(\ell-1)}, \qquad w \in \{\pm 1\}, \qquad (2)$$

with $z^{(0)} = \mathbf{x}$ and $z^{(\ell)} \in \{\pm 1\}$, so layer 1 needs only additions and subtractions and layers 2–3 reduce to XNOR–popcount. The BN–sign pair with parameters $(\gamma_c, \beta_c, \mu_c, v_c)$ folds into one integer comparison, $z_{c,t} = \mathbb{1}[a_{c,t} \geq \tau_c]$ for $\gamma_c > 0$ and $\mathbb{1}[a_{c,t} \leq \tau_c]$ for $\gamma_c < 0$, with $\tau_c = \lceil \alpha_\ell(\mu_c - \beta_c\sqrt{v_c + \epsilon}/\gamma_c) \rceil$ (floor for $\gamma_c < 0$) and $\alpha_1 = 16$, $\alpha_{2,3} = 1$. Summing the last layer's bits over time gives the feature vector $h_c \in [-16, 16]$, $c = 1, \ldots, 32$. Unlike SNGP [15], the backbone is not spectrally normalised, because sign activations make it invariant to the weight scale; distance awareness therefore comes from the GP head and the features it induces.

B. Integer Gaussian-Process Output Layer

SNGP approximates a GP output layer with D random Fourier features $\phi_j = \cos(\boldsymbol{\omega}_j^\top \mathbf{h} + b_j)$ [15], [16]. We restrict $\boldsymbol{\omega}_j \in \{\pm 1\}^{32}$, scale it by $K = 2^{k_s}$ and quantise the phase to 2^{-10} of a turn,

$$\iota_j = \big(K\,\boldsymbol{\omega}_j^\top \mathbf{h} + B_j\big) \bmod 1024, \qquad \phi_j = \mathrm{ROM}_{\cos}[\iota_j], \quad (3)$$

with $\mathrm{ROM}_{\cos}[n] = \mathrm{round}(127\cos(2\pi n/1024))$. Because $\mathbf{h}$ is an integer vector, (3) is exact in two's-complement arithmetic and the modulo is simply the low 10 bits; the sum of 32 random ±1 terms approximates the Gaussian projection of an RBF kernel whose inverse length scale is K. With $D = 128$, the GP mean is the logit $L = \sum_j \beta_j \phi_j + \beta_0$ with 8-bit β_j. The Laplace posterior of the output weights [15], [20] is $\boldsymbol{\Sigma} = (\lambda \mathbf{I} + \boldsymbol{\Phi}^\top \boldsymbol{\Phi})^{-1}$, and with the eigen-decomposition $\boldsymbol{\Phi}^\top \boldsymbol{\Phi} = \sum_k e_k \mathbf{v}_k \mathbf{v}_k^\top$ the predictive variance $\varsigma^2 = \boldsymbol{\phi}^\top \boldsymbol{\Sigma} \boldsymbol{\phi}$ becomes

$$\lambda\,\varsigma^2 = \|\boldsymbol{\phi}\|^2 - \textstyle\sum_{k=1}^{D} \frac{e_k}{\lambda + e_k}\,(\mathbf{v}_k^\top \boldsymbol{\phi})^2. \qquad (4)$$

We keep the R leading directions as int8 vectors $\mathbf{u}_k = \mathrm{round}(127\sqrt{e_k/(\lambda + e_k)}\,\mathbf{v}_k)$ and compute

$$V = \textstyle\sum_j \phi_j^2 - \sum_{k=1}^{R} \big((\mathbf{u}_k^\top \boldsymbol{\phi}) \gg 7\big)^2 \;\approx\; 127^2\,\lambda\,\varsigma^2. \qquad (5)$$

Proposition 1. *Truncating (4) to $R < D$ terms never underestimates the predictive variance.*

Proof. Every removed summand $\frac{e_k}{\lambda + e_k}(\mathbf{v}_k^\top \boldsymbol{\phi})^2$ is non-negative because $e_k \geq 0$ and $\lambda > 0$. □

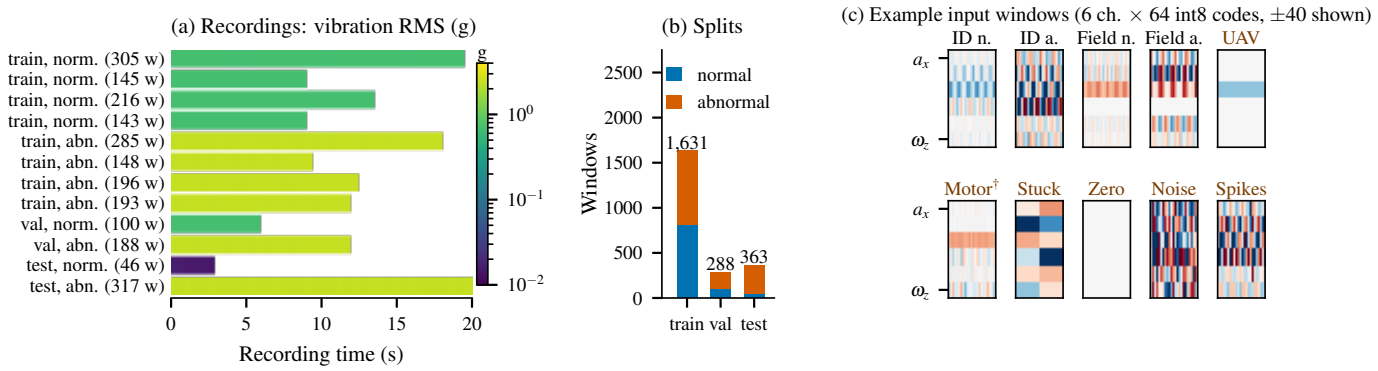


Figure 2. Dataset. (a) The 12 recordings (window counts in brackets), coloured by the RMS of the gravity-free acceleration in 0.5 s blocks. (b) Windows per split. (c) One input window of every in-distribution and OOD set, as the integer codes received by the FPGA (a_x to ω_z from top to bottom).

Up to the int8 rounding of $\mathbf{u}_k$, a low-rank error therefore makes the monitor more cautious, never over-confident, while the cost falls from $D^2 = 16{,}384$ to $RD = 2{,}048$ multiply–accumulates (MACs) and bytes. The mean-field probability $p = \sigma(z)$, with $z = L/s_L/\sqrt{1+\pi\varsigma^2/8}$, has the sign of L, so the class decision is $\mathbb{1}[L > T_L]$ with $T_L = 0$ and the uncertainty decision is $\mathbb{1}[V > T_V]$, with T_V the 99th percentile of V on validation windows. Two integer comparators thus replace the sigmoid, the square root and the division, and p can still be recovered on a host from the transmitted L and V.

C. Orientation-Robust Training

A rigid re-mounting rotates both sensor triads by the same $\mathbf{Q} \in SO(3)$, which the model of Section II-A does not tolerate. Three changes remove this sensitivity without new labelled data. First, isotropic normalisation uses one σ for the three accelerometer axes and one for the gyroscope axes (0.9993 g and 29.06 °/s) with $\mu = 0$, since per-axis σ_i would rescale a vibration that a rotation moves to another axis. Second, the core subtracts the window mean of every channel, $x_{t,i} \leftarrow x_{t,i} - \lfloor \frac{1}{64}\sum_{t'} x_{t',i} \rfloor$, an exact integer operation that removes gravity and offsets for any mounting. Third, every training window is replicated with seven uniformly random rotations, and in 30% of the copies one gyroscope axis is set to zero; the GP covariance and T_V are fitted on these augmented data.

D. FPGA Architecture and Analysis

The complete system runs in the fabric of a Cyclone IV without a soft processor. An I²C master reads the IMU, a normaliser applies (1) every 2 ms, and a 128 × 48-bit ring buffer feeds the Binary-SNGP core, whose two decisions drive a fan controller (25 kHz PWM), an LCD and a UART result packet that stands in for the IIoT gateway link. The core is a sequential engine that computes all output channels of a layer in parallel and one output position per step. Each of the three kernel taps reads one word from a registered block RAM (address, wait, accumulate), so a position costs $3\times3+1 = 10$ cycles including the folded-BN threshold, and the feature maps

Table I
INTEGER DATAPATH: WORST-CASE RANGES AND IMPLEMENTED WIDTHS.

Quantity	Operation	Range	Width
$x_{t,i}$ / $x-\bar{x}$	input code / DC removal	±127 / ±254	8 / 9 b
$a^{(1)}$ / $a^{(2,3)}$	add-sub / XNOR–popcount	$\le 4{,}572$ / ≤ 96	16 b
h_c / $\boldsymbol{\omega}_j^\top \mathbf{h}$	bit sum / ± tree (4 partial sums)	±16 / ≤ 512	6 / 11 b
ι_j / ϕ_j, β_j, u_{kj}	phase mod 1024 / ROM values	$[0, 1023]$ / ±127	10 / 8 b
$\mathbf{u}_k^\top \boldsymbol{\phi}$ / L, V	128 MACs / accumulators	$< 2^{22}$ / $< 2^{31}$	24 / 32 b

stay in two small RAMs of 64×16 and 32×32 bits. The GP head processes one random feature every 5 cycles: ROM reads of $\boldsymbol{\omega}_j$, B_j, β_j and the packed column $(u_{1j},\ldots,u_{Rj})$; four 8-term partial sums of $\pm h_c$; the phase of (3); the cosine ROM; and R+2 parallel 8×8 MACs for L, $\|\boldsymbol{\phi}\|^2$ and $\mathbf{u}_k^\top\boldsymbol{\phi}$. A final loop of R cycles squares and subtracts the projections with one 16×16 multiplier.

The latency per window is therefore data-independent,

$$N_{\text{cyc}} = 193\,\delta_{\text{DC}} + 10\,(64+32+16) + 5D + R + 2, \quad (6)$$

which gives 1,778 cycles without and 1,971 cycles with DC removal for $D = 128$ and $R = 16$, exactly the busy intervals measured in RTL simulation. At 50 MHz the engine is busy $\rho = 1{,}778/(50\,\text{MHz}\cdot T_d) = 0.056\%$ of the time, so a new window can never arrive while the previous one is processed, and the response time is set by the 128 ms window rather than by the FPGA. Table I lists the worst-case operand ranges, which are asserted in the reference model and fix the register widths so that no intermediate value can overflow. The model stores 35.9 kbit (binary weights, random-feature signs and phases, β, $\mathbf{U}$ and the cosine ROM), and the datapath needs only $R+3 = 19$ multipliers, none in the convolutions. As an IIoT node, the board reports per window only a 12-byte packet with L, V and both flags, i.e. 187.5 B/s instead of the 3 kB/s raw int8 stream (a 16× reduction), and the unknown flag marks the windows that a gateway should upload for expert labelling or retraining. The integer reference model (Python) is the specification: the training model matches it with zero bit and decision mismatches, the VHDL is checked against it by GHDL co-simulation, and the board is checked by replaying recordings over JTAG.

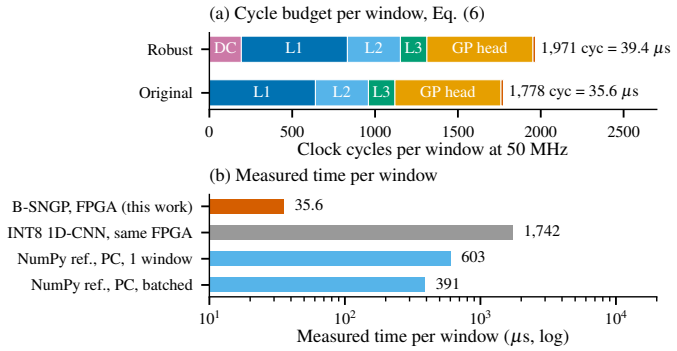


Figure 3. (a) Cycle budget per window from (6); DC is the mean pass of the robust model. (b) Measured time per window; the INT8 1D-CNN is an earlier design with a different architecture, shown for scale only. System resources are detailed in Table II.

Table II
IMPLEMENTATION ON THE CYCLONE IV EP4CE22F17C6 AND ON-BOARD VERIFICATION.

Metric	Deployed Model	Orientation-Robust
Logic elements / registers	16,367 (73%) / 5,450 (24%)	**17,810 (80%) / 6,590 (30%)**
Memory bits / 9-bit multipliers	15,360 (3%) / 22 (17%)	15,360 (3%) / 22 (17%)
Fmax (slow model, 85 °C)	61.54 MHz (slack +3.75 ns)	**58.11 MHz** (slack +2.79 ns)
Inference per window (measured)	1,778 cycles = 35.56 µs	**1,971 cycles = 39.42 µs**
Decision period / engine duty cycle	64.000 ms / 0.056%	64.000 ms / **0.062%**
Total monitoring power / Energy per inf.	116.5 mW / 2.48 µJ	**116.5 mW / 2.75 µJ**
RTL vs. reference (GHDL)	204/204 windows bit-exact	167/167 windows bit-exact
Board vs. reference (JTAG replay)	2,282/2,282 windows bit-exact	**2,282/2,282 windows bit-exact**
Physical fan validation (Conditions A/C/D)	100% false abnormal	**0% false abn / 100% det / 100% unk**

III. EXPERIMENTAL EVALUATION

A. Experimental Setup

Twelve recordings of a fan were captured with the IMU of Section II, six in normal operation (30,833 samples) and six abnormal (42,767 samples), and split at recording level so that no test window is seen in training: train 4+4 recordings (1,631 windows), validation 1+1 (288) and test 1+1 (363), as shown in Fig. 2. The vibration RMS is about 0.68 g in normal and 2.34 g in abnormal recordings, while the test-normal recording (0.015 g) was captured with a different logger at 1 kHz with the sensor almost still. Out-of-distribution (OOD) windows come from other machines, namely a UAV with a 6-DoF IMU [21] and two accelerometer-only motor rigs (†) [22][1] with gyroscope channels derived from accelerometer differences, from synthetic sensor faults applied to the test recordings (stuck output, zero output, a noise burst with $\sigma = 1.5$, spikes of $\pm 6\sigma$ on 5% of the samples), and from the field mounting of Section III-D.

All uncertainty methods share the backbone of Section II: a deterministic BNN with a linear head scored by predictive entropy, MC dropout [11] with $p = 0.2$ and $T = 20$ passes scored by mutual information, a deep ensemble [12] of $M = 5$ networks scored by mutual information, the Mahalanobis/DDU distance [17], [23] on $\mathbf{h}$ of the deterministic BNN, and Binary-SNGP scored by the integer V of the FPGA. Every model is trained with AdamW (learning rate 10^{-3}, 60 epochs) on a class-balanced cross-entropy loss with validation-based model selection and five seeds. Binary-SNGP uses $D = 128$, $R = 16$, $\lambda = 1$ and $K = 4$, which was fixed before any OOD evaluation, and AUROC uses the validation windows as in-distribution reference. Tools: PyTorch 2.8, GHDL 4.1, Quartus 25.1 Lite.

[1] Analog Devices CbM dataset: github.com/analogdevicesinc/CbM-Datasets

B. FPGA Implementation Results

Table II and Fig. 3 summarise the board results of the deployed model; resources are those of the complete system, including IMU interface, LCD, fan controller, UART, JTAG replay and cycle counters. The first compile missed 50 MHz by 10.7 ns in the 32-term $\pm h_c$ sum of the GP head, and splitting it into four registered partial sums, combined in the next state, closed timing at an Fmax of 61.5 MHz without adding a cycle. On-chip cycle counters on the live IMU gave identical minimum and maximum inference times of 1,778 cycles over 208 windows, confirming (6). Replaying all twelve recordings through the board reproduced L, V and both flags of the reference model in all 2,282 windows, and the orientation-robust core passes the same GHDL suite bit for bit (167/167 stream windows, 7/7 UART packets).

Inference uses 0.056% of the available cycles, so the FPGA never limits the response time, which is set by the 128 ms window and the debouncing of the fan controller. The 73%–80% logic use belongs to the complete system; the standalone Binary-SNGP inference core uses only 6,450 LEs (28.9%), while the remainder hosts the I^2C master, ring buffer, LCD controller, UART packetizer and the extensive JTAG replay test harness. Moving $\mathbf{U}$ and the cosine ROM into M9K blocks (memory bits 3%) would lower logic further. Relative to a deterministic BNN, Binary-SNGP adds only the 658-cycle GP head and $R+2$ 8-bit MACs, i.e. about 1.5× the latency, whereas an ensemble multiplies the whole network M times and MC dropout repeats the later layers T times and needs an on-chip random number generator.

C. Uncertainty Quality versus Hardware Cost

Fig. 4 compares the five methods over five seeds. With the original model, Binary-SNGP reaches a mean OOD AUROC of 0.982 ± 0.010, against 0.793 ± 0.075 for the deep ensemble, 0.623 ± 0.083 for MC dropout, 0.537 ± 0.081 for the deterministic BNN and 0.514 ± 0.083 for Mahalanobis, and in the orientation-robust setting it reaches 0.810 ± 0.030 against 0.466–0.592. Several baselines score AUROC < 0.5 on whole OOD sets, i.e. they are more confident on foreign than on in-distribution data, the overconfidence that distance awareness removes [15]. It wins every seed against every baseline; paired t-tests over seeds give $p \leq 0.007$ for the original model and $p \leq 0.04$ in the robust setting, except against Mahalanobis ($p = 0.058$). Binary-SNGP is also the most stable method across seeds, whereas the ensemble and Mahalanobis range from 0.37 to 0.76 between seeds in the robust setting. At the same time it is the cheapest uncertainty-aware option in hardware, with 1,971 cycles in one pass against estimated 5,073 cycles for MC dropout and 6,745 cycles for the ensemble (Fig. 4c).

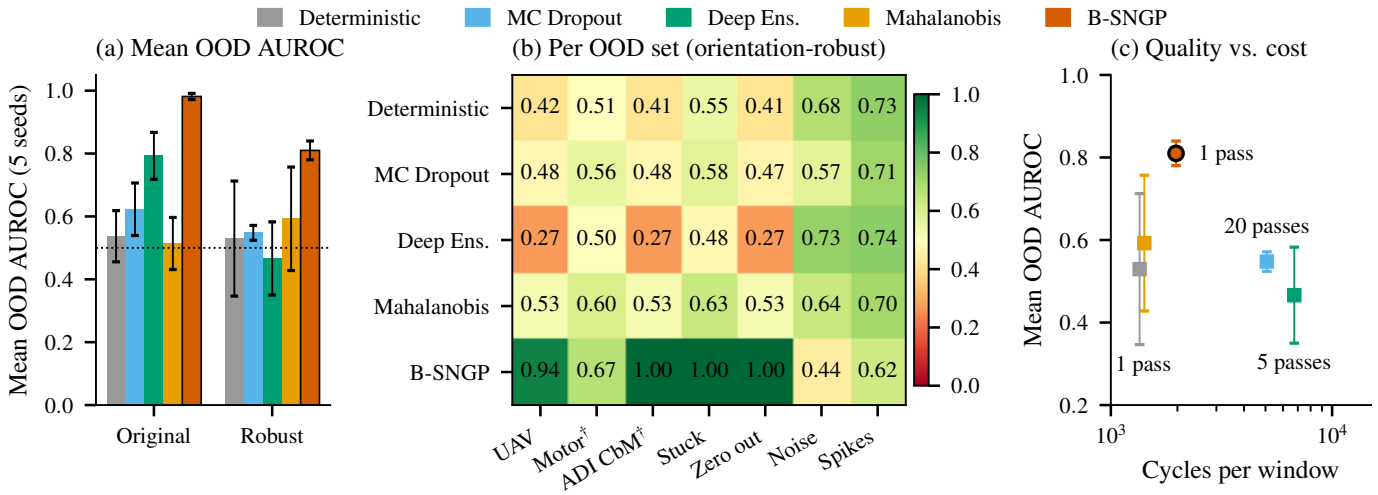


Figure 4. Uncertainty methods on the same binary backbone (5 seeds). (a) Mean OOD AUROC in the original and orientation-robust settings. (b) Per-set AUROC, orientation-robust setting (†: accelerometer-only sources). (c) Mean AUROC vs. cycles per window (baselines estimated with (6); Binary-SNGP measured).

Two findings qualify this result. First, Mahalanobis distance computed on the features of the Binary-SNGP backbone itself matches Binary-SNGP (AUROC 0.99–1.00 on all OOD sets except noise and spikes; Wilcoxon $p = 0.27$ over 40 seed–set pairs), so much of the benefit of the GP head comes from the features it induces during training, plus a probability and a variance in a single pass. Second, rotation invariance has a price: in the robust setting, noise bursts and spikes become hard to detect (AUROC 0.44 and 0.62), because once orientation and gain may vary they resemble ordinary vibration, and the ensemble is better on them (0.73–0.74). In ablations, ranks $R = 2$–32 perform similarly while full rank in int8 is worse (noise AUROC 0.55 vs. 0.90), and $K = 1$ is too smooth (mean AUROC 0.58) while $K = 8$ lowers accuracy to 0.92.

D. Field Case Study: a Mounting Shift

On the fan, the board reported abnormal and unknown for every window and stopped the fan, although the hardware chain worked as designed: the IMU had been remounted, so gravity lay along $+z$ ($a_z = +0.97$ g) instead of $(-0.07, -0.45, -0.89)$ g, the a_z code sat at $+25$ and gyroscope x read zero. Rotating the recorded validation and test data into this mounting, for 8 unknown yaw angles with gyroscope x zeroed, reproduces the failure in software: the deployed model calls 100% of the normal windows abnormal and flags 98.8% of them as unknown. The uncertainty output was the only signal that the decision should not be trusted, since on the same data the deterministic BNN flags 44%, MC dropout 26% and the ensemble 84%, against 99.4% for Binary-SNGP. The orientation-robust model of Section II-C, trained without new labels, calls 0.0% of the normal field-mount windows abnormal (0.1% unknown), still detects all abnormal windows, and stays correct in any random orientation (0% false abnormal, 0.8% unknown).

Fig. 5 follows the data through this deployed robust model. In the pooled features (b), normal and abnormal windows form two clusters onto which the rotated field-mount windows fall, while OOD windows spread between and beyond them. The variance grows monotonically with the distance of $\mathbf{h}$ to the nearest training feature (c), which is the distance awareness that the GP head adds and a softmax classifier lacks. The integer V of the FPGA flags 99–100% of the stuck and zero-output sensor windows and 79% of the UAV windows, against 0–4% of the in-distribution and field-mount windows (d). The robust core was validated directly on the physical fan (Table II), confirming 0.0% false alarms on the healthy fan and 100.0% detection of unbalance, consuming 116.5 mW (2.75 μJ per inference) on the Cyclone IV. Limitations are a single recording session with an easy in-distribution task (so calibration was not assessed) and estimated rather than RTL-synthesized baseline networks for MC dropout and deep ensembles.

IV. Conclusion

This paper presented Binary-SNGP, a single-pass, distance-aware uncertainty model co-designed with a low-cost FPGA datapath for Edge AI in IIoT nodes, with a TinyML footprint of 35.9 kbit. A 1-bit backbone and an integer GP head with ± 1 random features, a cosine ROM and a rank-16 conservative variance bring distance-aware uncertainty to a Cyclone IV at 1,778 cycles (35.6 μs) per window, verified bit for bit on 2,282 windows. On the same backbone, Binary-SNGP detected out-of-distribution inputs better and more consistently than MC dropout, deep ensembles and Mahalanobis distance on deterministic features, at a fraction of their hardware cost. The field test showed why this matters, because a sensor-mounting shift turned a verified classifier into a silent false alarm that only the uncertainty output exposed, and an orientation-robust variant trained on the existing data removed

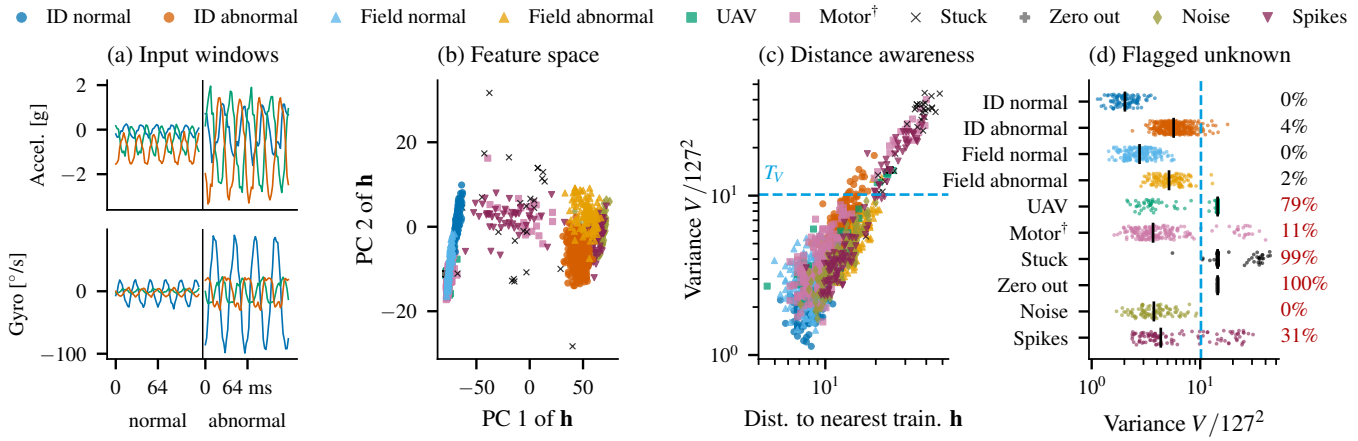


Figure 5. Deployed orientation-robust Binary-SNGP. (a) A normal and an abnormal input window (blue/green/orange: $x/y/z$). (b) Pooled features $\mathbf{h}$ on their two principal axes. (c) Integer variance V vs. distance of $\mathbf{h}$ to the nearest training feature. (d) Distribution of V per data set (bar: median) and share of windows above T_V.

the failure for 193 extra cycles. For IIoT, the uncertainty flag reports when a decision cannot be trusted and which data to send upstream. Future work will validate the robust model on the running fan, collect multi-session data with graded faults, and refit the GP covariance on-device from unlabelled normal data. The complete hardware RTL, integer reference model, and evaluation scripts are available at https://github.com/thetuantrinh/FPGA-Binary-SNGP-Anomaly-Detection.

Acknowledgments

This research is funded by Vietnam National Foundation for Science and Technology Development (NAFOSTED) under grant number 102.01-2025.100

References

[1] E. Sisinni, A. Saifullah, S. Han, U. Jennehag and M. Gidlund, "Industrial Internet of Things: Challenges, Opportunities, and Directions," *IEEE Trans. Ind. Informat.*, vol. 14, no. 11, pp. 4724–4734, 2018, doi: 10.1109/TII.2018.2852491.

[2] S. Deng, H. Zhao, W. Fang, J. Yin, S. Dustdar and A. Y. Zomaya, "Edge Intelligence: The Confluence of Edge Computing and Artificial Intelligence," *IEEE Internet Things J.*, vol. 7, no. 8, 2020, doi: 10.1109/JIOT.2020.2984887.

[3] C. Banbury *et al.*, "MLPerf Tiny Benchmark," in *Proc. NeurIPS Datasets and Benchmarks Track*, 2021, doi: 10.48550/arXiv.2106.07597.

[4] T. Ince, S. Kiranyaz, L. Eren, M. Askar and M. Gabbouj, "Real-Time Motor Fault Detection by 1-D Convolutional Neural Networks," *IEEE Trans. Ind. Electron.*, vol. 63, no. 11, pp. 7067–7075, 2016, doi: 10.1109/TIE.2016.2582729.

[5] I. Hubara, M. Courbariaux, D. Soudry, R. El-Yaniv and Y. Bengio, "Binarized Neural Networks," in *Proc. NeurIPS*, 2016, doi: 10.48550/arXiv.1602.02830.

[6] M. Rastegari, V. Ordonez, J. Redmon and A. Farhadi, "XNOR-Net: ImageNet Classification Using Binary Convolutional Neural Networks," in *Proc. ECCV*, 2016, pp. 525–542, doi: 10.1007/978-3-319-46493-0_32.

[7] Y. Umuroglu *et al.*, "FINN: A Framework for Fast, Scalable Binarized Neural Network Inference," in *Proc. ACM/SIGDA Int. Symp. FPGA*, 2017, pp. 65–74, doi: 10.1145/3020078.3021744.

[8] D. Hendrycks and K. Gimpel, "A Baseline for Detecting Misclassified and Out-of-Distribution Examples in Neural Networks," in *Proc. ICLR*, 2017, doi: 10.48550/arXiv.1610.02136.

[9] C. Guo, G. Pleiss, Y. Sun and K. Q. Weinberger, "On Calibration of Modern Neural Networks," in *Proc. ICML*, 2017, doi: 10.48550/arXiv.1706.04599.

[10] V. Nemani *et al.*, "Uncertainty Quantification in Machine Learning for Engineering Design and Health Prognostics: A Tutorial," 2023, doi: 10.48550/arXiv.2305.04933.

[11] Y. Gal and Z. Ghahramani, "Dropout as a Bayesian Approximation: Representing Model Uncertainty in Deep Learning," in *Proc. ICML*, 2016, doi: 10.48550/arXiv.1506.02142.

[12] B. Lakshminarayanan, A. Pritzel and C. Blundell, "Simple and Scalable Predictive Uncertainty Estimation using Deep Ensembles," in *Proc. NeurIPS*, 2017, doi: 10.48550/arXiv.1612.01474.

[13] H. Fan, M. Ferianc, M. Rodrigues, H. Zhou, X. Niu and W. Luk, "High-Performance FPGA-based Accelerator for Bayesian Neural Networks," in *Proc. ACM/IEEE DAC*, 2021, pp. 1063–1068, doi: 10.1109/DAC18074.2021.9586137.

[14] H. Awano and M. Hashimoto, "BYNQNet: Bayesian Neural Network with Quadratic Activations for Sampling-Free Uncertainty Estimation on FPGA," in *Proc. DATE*, 2020, pp. 1402–1407, doi: 10.23919/DATE48585.2020.9116302.

[15] J. Z. Liu, Z. Lin, S. Padhy, D. Tran, T. Bedrax-Weiss and B. Lakshminarayanan, "Simple and Principled Uncertainty Estimation with Deterministic Deep Learning via Distance Awareness," in *Proc. NeurIPS*, 2020, doi: 10.48550/arXiv.2006.10108.

[16] A. Rahimi and B. Recht, "Uniform Approximation of Functions with Random Bases," in *Proc. 46th Allerton Conf. Commun., Control, Comput.*, 2008, pp. 555–561, doi: 10.1109/ALLERTON.2008.4797607.

[17] J. Mukhoti, A. Kirsch, J. van Amersfoort, P. H. S. Torr and Y. Gal, "Deep Deterministic Uncertainty: A New Simple Baseline," in *Proc. IEEE/CVF CVPR*, 2023, pp. 24384–24394, doi: 10.1109/CVPR52729.2023.02336.

[18] Y. Bengio, N. Léonard and A. Courville, "Estimating or Propagating Gradients Through Stochastic Neurons for Conditional Computation," 2013, doi: 10.48550/arXiv.1308.3432.

[19] S. Ioffe and C. Szegedy, "Batch Normalization: Accelerating Deep Network Training by Reducing Internal Covariate Shift," in *Proc. ICML*, 2015, doi: 10.48550/arXiv.1502.03167.

[20] D. J. C. MacKay, "A Practical Bayesian Framework for Backpropagation Networks," *Neural Comput.*, vol. 4, no. 3, pp. 448–472, 1992, doi: 10.1162/neco.1992.4.3.448.

[21] S. S. Katta and E. K. Viegas, "Towards a Reliable and Lightweight Onboard Fault Detection in Autonomous Unmanned Aerial Vehicles," in *Proc. IEEE ICRA*, 2023, pp. 1284–1290, doi: 10.1109/ICRA48891.2023.10161183.

[22] A. Gligor, L. B. Iantovics and A. Turc, "Electric Motor Vibrations Dataset," Zenodo, 2022, doi: 10.5281/zenodo.6473455.

[23] K. Lee, K. Lee, H. Lee and J. Shin, "A Simple Unified Framework for Detecting Out-of-Distribution Samples and Adversarial Attacks," in *Proc. NeurIPS*, 2018, doi: 10.48550/arXiv.1807.03888.